Name: _____

Making Ionic Compounds

1. Heat some magnesium ribbon over a Bunsen burner. Do not look at the flames. **WEAR SAFETY GLASSES.**

What is the symbol for Magnesium? _____

What is the symbol for Oxygen? _____

What colour was the material produced? _____

The material produced is Magnesium oxide.

When the Magnesium burns each magnesium atom loses 2 electrons and forms the ion

Mg^{+2}

The oxygen atoms in the air near the burning magnesium gain two electrons and form the ion

O^{-2}

These ions have different charges (positive and negative) so they will join together with _____ bonds.

Write the formula for the magnesium oxide.

2. Sprinkle some iron filings over the blue heating flame of a Bunsen burner. **WEAR SAFETY GLASSES.**

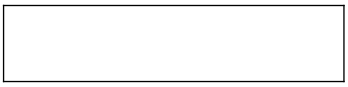
What is the symbol for iron? _____

What colour was the material produced? _____

The burning causes the iron to mix with the oxygen. What is the name of the substance produced?

_____

When the iron burns each iron atom loses 3 electrons and forms the ion



The oxygen atoms in the air near the burning iron gain two electrons and form the ion



These ions have different charges (positive and negative) so the will bond together to form _____ bonds.

Write the formula for the iron oxide.

3. Heat some fine copper wire over a Bunsen burner. **WEAR SAFETY GLASSES.**

What is the symbol for copper? _____

What colour was the material produced? _____

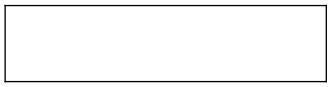
What is the name of the material produced?

_____

Copper is a metal, so when the copper burns each copper atom _____ 2 electrons and forms the ion:



Oxygen is a _____ _____, so oxygen atoms in the air near the burning copper _____ two electrons and form the ion:



These ions have different charges (copper is _____ and oxygen _____) so they will bond together to form _____ bonds.

Write the formula for the copper oxide.

Table of common ions

Cations		Anions	
+1 charge		**- 1 charge**	
Hydrogen	H^+	Fluoride	F^-
Lithium	Li^+	Chloride	Cl^-
Sodium	Na^+	Bromide	Br^-
Potassium	K^+	Iodide	I^-
Copper (I)	Cu^+	Hydride	H^-
Silver	Ag^+	Hydroxide	OH^-
Ammonium	NH_4^+	Nitrite	NO_2^-
		Nitrate	NO_3^-
+2 charge		**- 2 charge**	
Manganese	Mn^{2+}	Oxide	O^{2-}
Magnesium	Mg^{2+}	Sulfide	S^{2-}
Calcium	Ca^{2+}	Carbonate	CO_3^{2-}
Barium	Ba^{2+}	Sulfate	SO_4^{2-}
Zinc	Zn^{2+}	Sulfite	SO_3^{2-}
Copper (II)	Cu^{2+}		
Mercury (II)	Hg^{2+}		
Iron (II)	Fe^{2+}		
Tin (II)	Sn^{2+}		
Lead (II)	Pb^{2+}		
Nickel (II)	Ni^{2+}		
Beryllium	Be^{2+}		
+3 charge		**- 3 charge**	
Aluminium	Al^{3+}	Nitride	N^{3-}
Iron (III)	Fe^{3+}	Phosphate	PO_4^{3-}
Chromium	Cr^{3+}	Phosphide	P^{3-}